

Energy Formuli & Conservation of Energy

Name:

Date:

A. Lesson Recap

- 1) Energy is the ability to do _____.
- 2) Mechanical energy is energy associated with _____.
- 3) List the three forms of mechanical energy:

_____, _____, _____

- 4) Why do we say “*at that instant*” when identifying energy?

B. The Actual Formulas

We will now calculate energy values using real numbers.

- 1) **Kinetic Energy** is given by the following formula:

This formula relates to the energy an object gives off in motion.

- 2) **Gravitational Potential Energy** is given by the following formula:

This formula relates to the energy an object could give off when at rest.

- 3) **Elastic Potential Energy**

This formula relates to the energy an object could give off when a spring stretched or compressed.

C. Mechanical Energy vs Total Energy Review

Mechanical Energy includes: KE , PE_g , PE_s

Mechanical energy assumes no energy loss (frictionless surfaces, no air resistance).

Total Energy includes: KE , PE_g , PE_s **and** thermal energy E_{th}

When any internal energy is present (friction, air resistance, heat), some mechanical energy becomes thermal energy, so mechanical energy is **not conserved**, but total energy still is (in a closed system).

D. Conservation of Energy Review

In a closed system:

In an open system:

What is the conservation of mechanical energy formula?

What is the conservation of total energy formula?

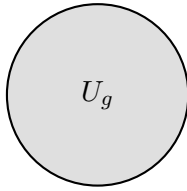
E. Pie Charts Review

Energy pie charts represent different instances of total energy distributions.

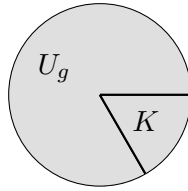
If mechanical energy is conserved, the **total size** of the mechanical-energy pie stays the same. The **slices/distributions** change.

If friction/heat/air resistance is present, the **mechanical pie shrinks**, and the missing part appears as thermal energy in a total energy model.

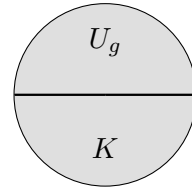
Example 1:



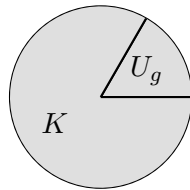
Instant 1: entirely U_g



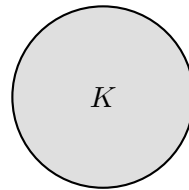
Instant 2: mostly U_g



Instant 3: half U_g , half K



Instant 4: mostly K

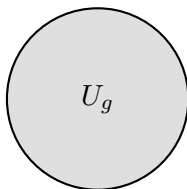


Instant 5: entirely K

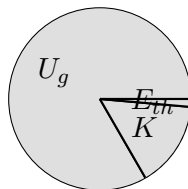
What energy is conserved in these pie chart scenario?

What energy is not conserved in these pie chart scenario?

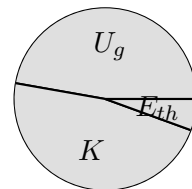
Example 2:



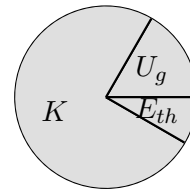
Instant 1: entirely U_g



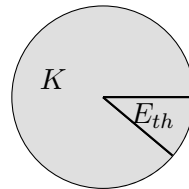
Instant 2: mostly U_g



Instant 3: half U_g , half K



Instant 4: mostly K



Instant 4: mostly K

What energy is conserved in these pie chart scenario?

What energy is not conserved in these pie chart scenario?

F. Mechanical Energy Conservation Examples

Example 1: Dropped object (no air resistance)

A 2.0 kg ball is dropped from rest from a height of 10 m.

Define the system: _____

What is the reference point ($h = 0$): _____

Energies at instants: Release ($h = 10$ m), Just after release ($h \approx 9.9$ m), Midway ($h = 5$ m), Just before impact ($h \approx 0.1$). At impact ($h = 0$ m).

(a) Find the energy values at release:

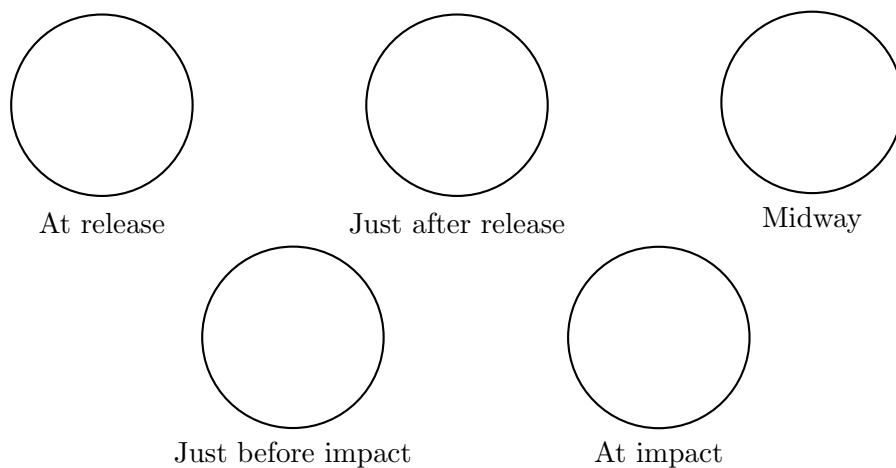
(b) Find the energy values just after release:

(c) Find the energy values midway down:

(d) Find the energy values just before impact:

(e) Find the energy values at impact:

(f) Draw the energy pie charts:



(g) Is mechanical energy conserved?

(h) Is total energy conserved?

Example 2: Skateboarder hill (frictionless)

A 60 kg skateboarder starts from rest at the top of a hill of height 5.0 m and rolls down smoothly.

Define the system: _____

What is the reference point ($h = 0$): _____

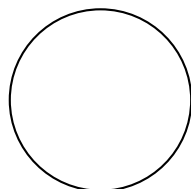
Energies at instants: At the top ($h = 5$ m), At the bottom ($h = 0$ m).

(a) Find the energy values at the top.

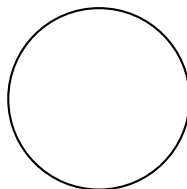
(b) Find the energy values at the bottom.

(c) Find the speed at the bottom.

(d) Draw the energy pie charts:



Top of hill



Bottom of hill

(e) Is mechanical energy conserved?

(f) Is total energy conserved?

Example 3: Spring launcher (frictionless)

A cart of mass 0.50 kg is launched by a spring with $k = 200$ N/m. The spring is compressed $x = 0.20$ m and released on a frictionless track.

Define the system: _____

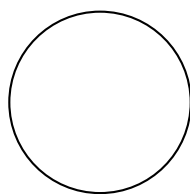
What is the reference point ($h = 0$): _____

Energies at instants: At release (compressed), After released (relaxed).

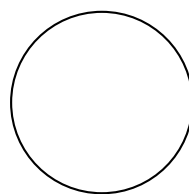
(a) Find the energy values at release.

(b) Find the energy values after release.

(c) Draw the energy pie charts:



Spring compressed



Cart moving

(d) Is mechanical energy conserved?

(e) Is total energy conserved?

G. Total Energy Conservation Examples

Example 1: Dropped object (with air resistance)

A 2.0 kg ball is dropped from rest from a height of 10 m. Due to air resistance, the ball hits the ground with speed $v_{\text{impact}} = 12 \text{ m/s}$.

Define the system: _____

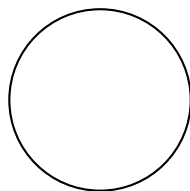
What is the reference point ($h = 0$): _____

Energies at instants: Release ($h = 10 \text{ m}$), Midway ($h = 5 \text{ m}$), At impact ($h = 0 \text{ m}$).

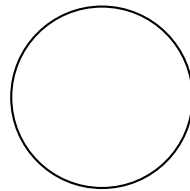
(a) Find the energy values at release.

(b) Find the energy values at impact ($h = 0 \text{ m}$).

(c) Draw the energy pie charts:



Release



Impact

(d) Is mechanical energy conserved?

(e) Is total energy conserved?

Example 2: Skateboarder hill (with friction)

A 60 kg skateboarder starts from rest at the top of a hill of height 5.0 m. Because of friction, the skateboarder reaches the bottom with speed $v = 8.0 \text{ m/s}$.

Define the system: _____

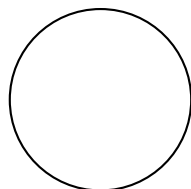
What is the reference point ($h = 0$): _____

Energies at instants: At the top ($h = 5 \text{ m}$), At the bottom ($h = 0 \text{ m}$).

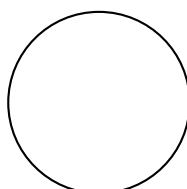
(a) Find the energy values at the top.

(b) Find the energy values at the bottom.

(c) Draw the energy pie charts:



Top of hill



Bottom of hill

(d) Is mechanical energy conserved?

(e) Is total energy conserved?

Example 3: Spring launcher (with friction)

A cart of mass 0.50 kg is launched by a spring with $k = 200\text{ N/m}$. The spring is compressed $x = 0.20\text{ m}$ and released. Because of friction, not all spring energy becomes kinetic energy. The cart leaves the spring region with speed $v = 3.0\text{ m/s}$.

Define the system: _____

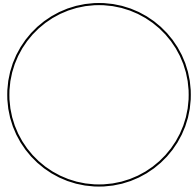
What is the reference point ($h = 0$): _____

Energies at instants: At release (compressed), After released (spring relaxed).

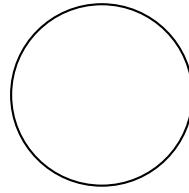
(a) Find the energy values at release.

(b) Find the energy values after release (spring relaxed).

(c) Draw the energy pie charts:



Spring compressed



Cart moving

(d) Is mechanical energy conserved?

(e) Is total energy conserved?